

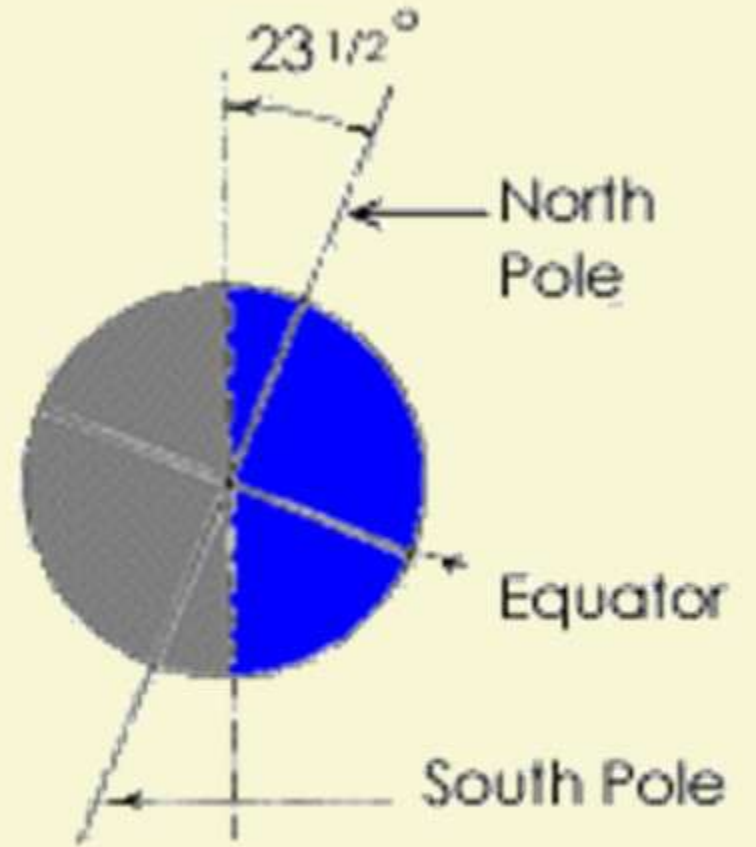
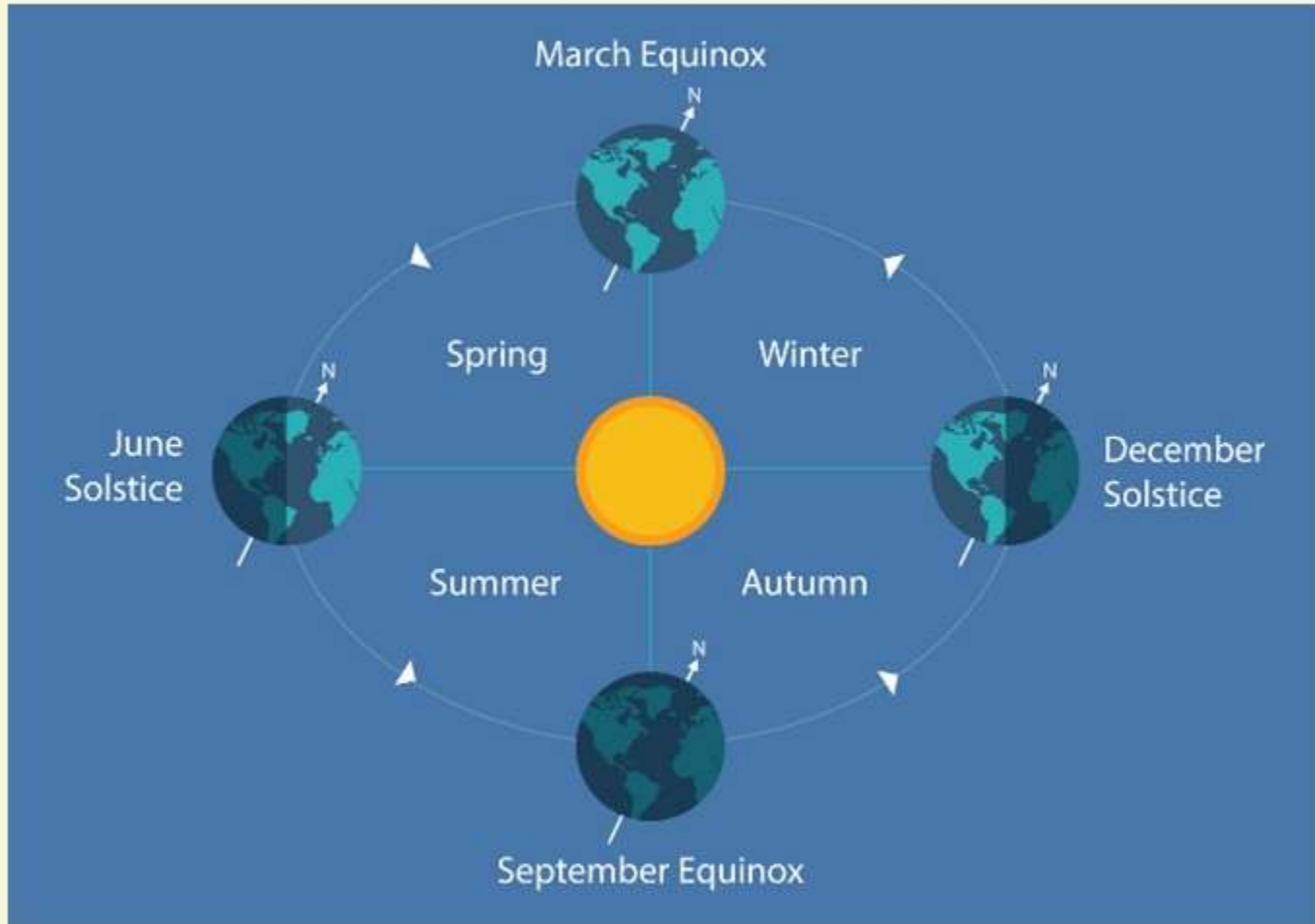
GEOGRAPHY

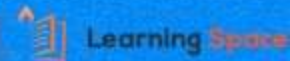
PHYSICAL GEOGRAPHY



MODULE - 8

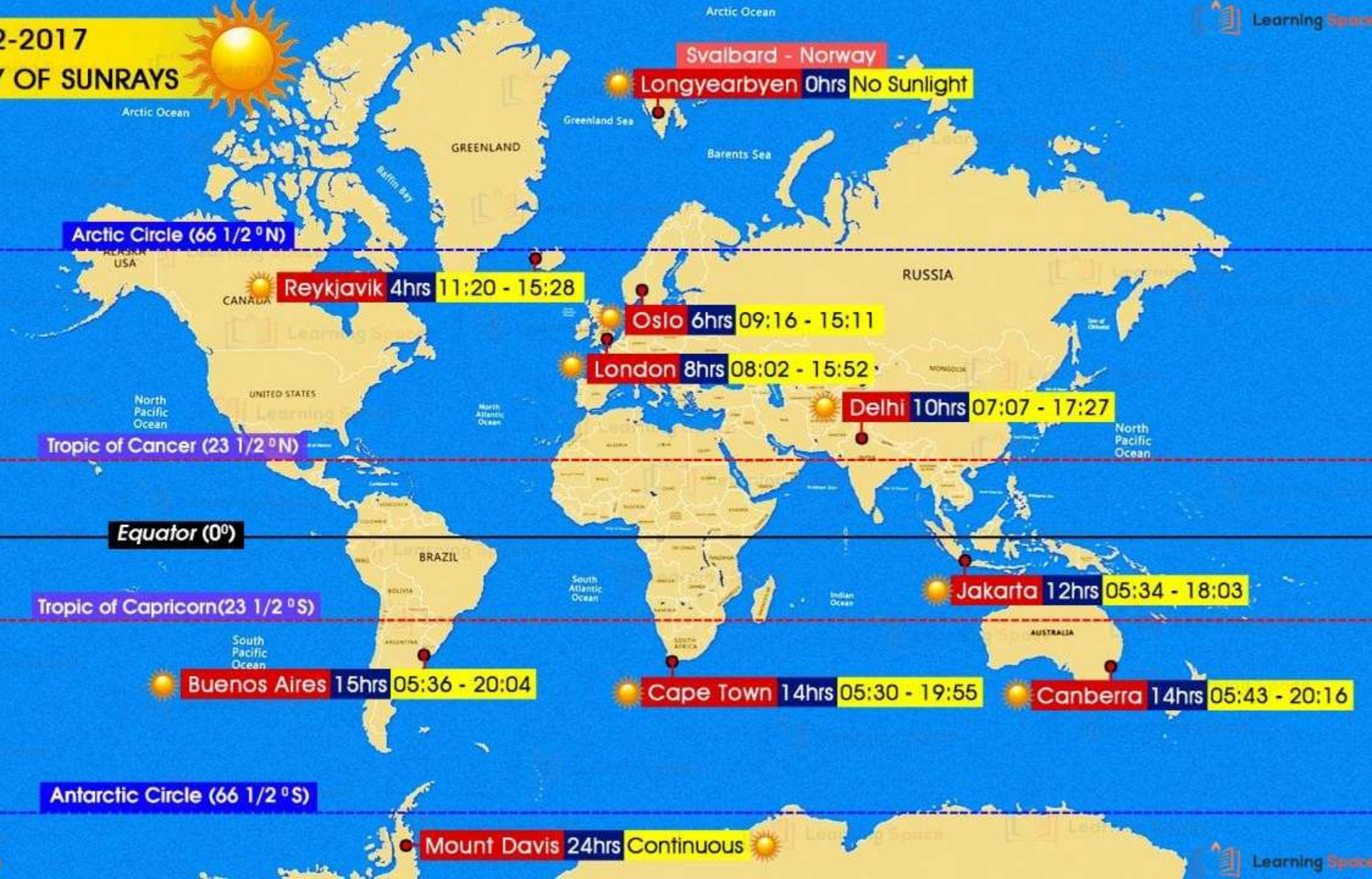
- VARIATIONS OF SUNLIGHT
- AXIAL TILT AND WHAT HAPPENS?
- EARTH'S ROTATION AND REVOLUTION
- SEASONS





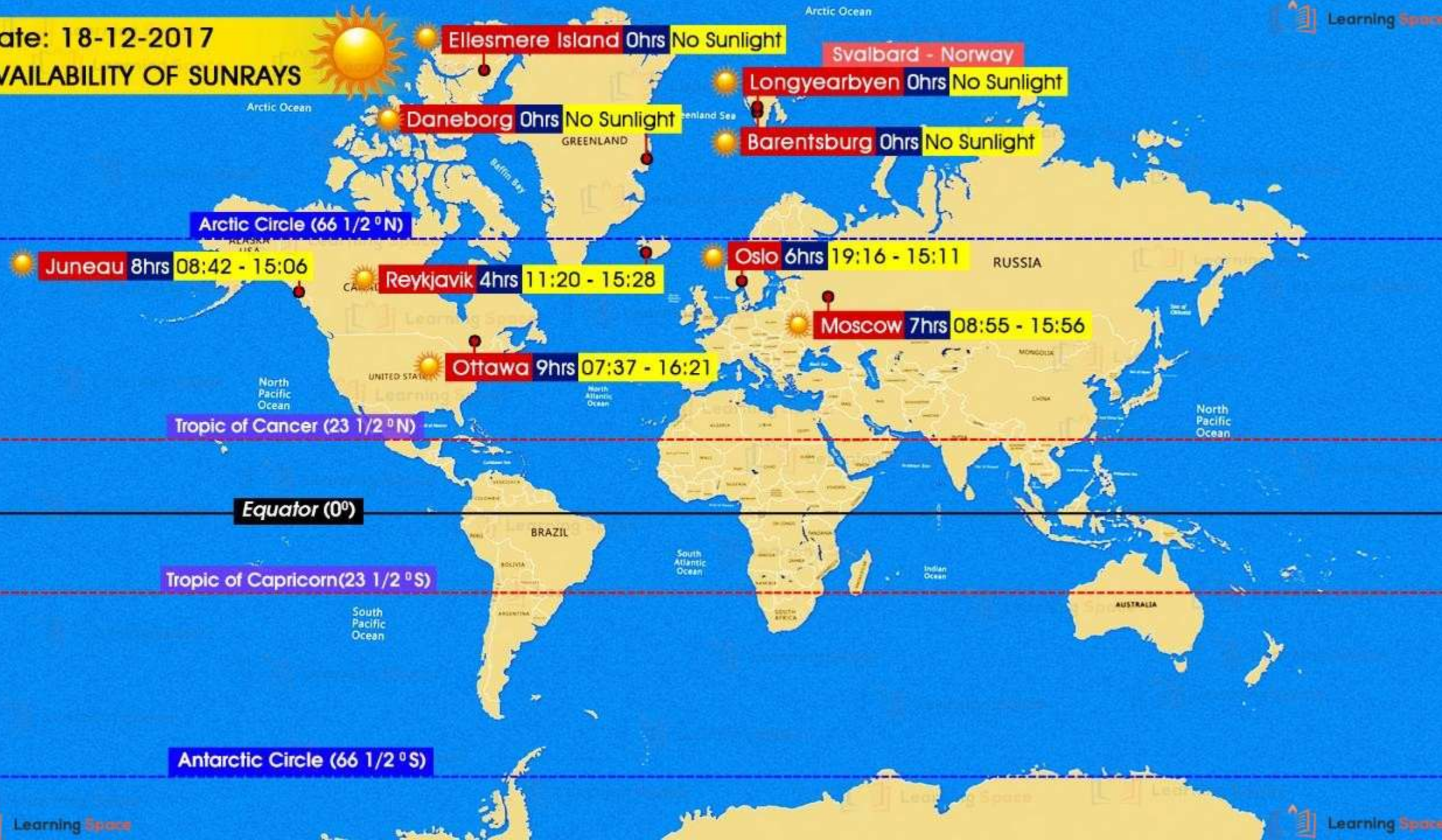
Date: 18-12-2017

AVAILABILITY OF SUNRAYS



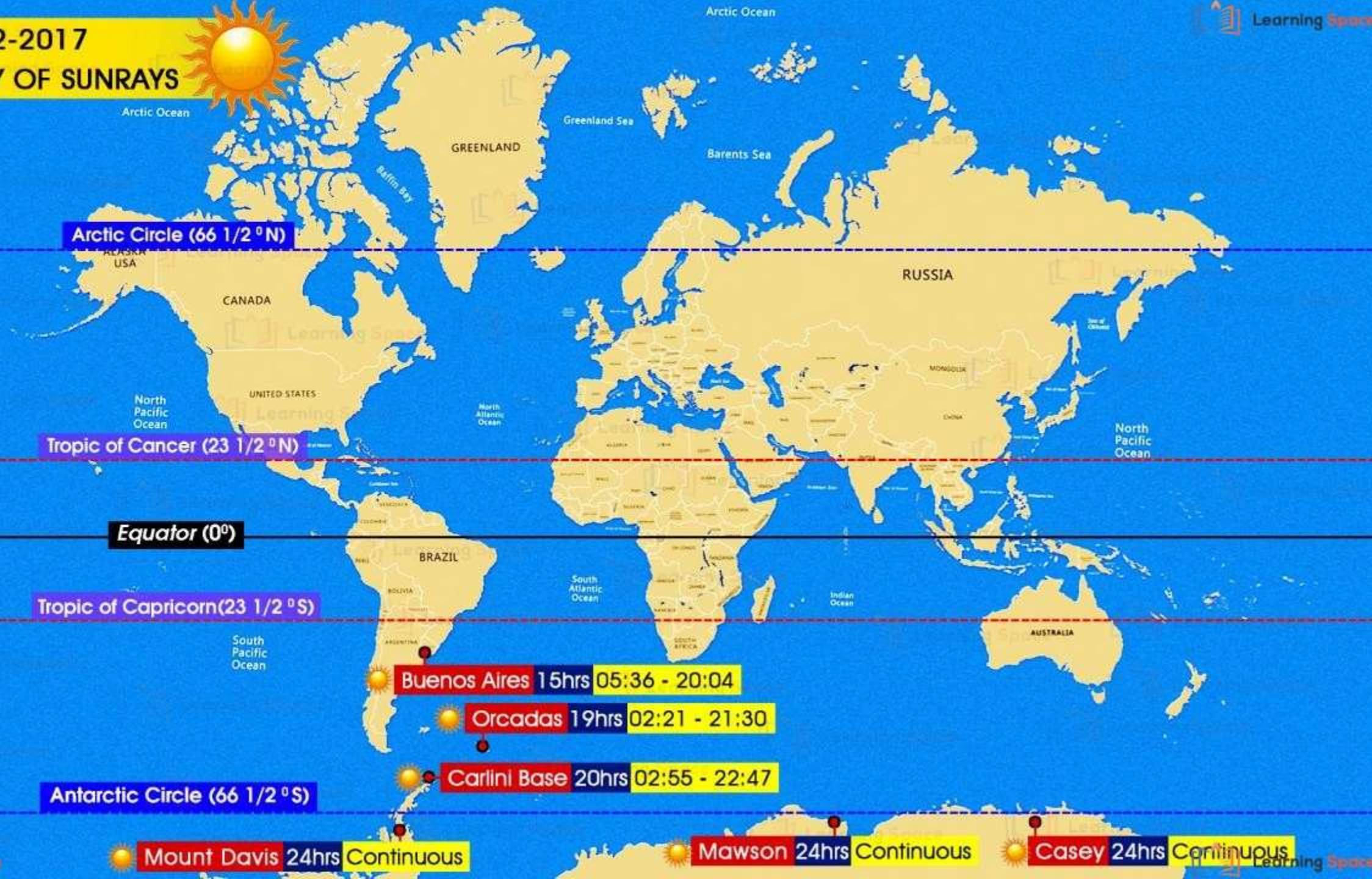
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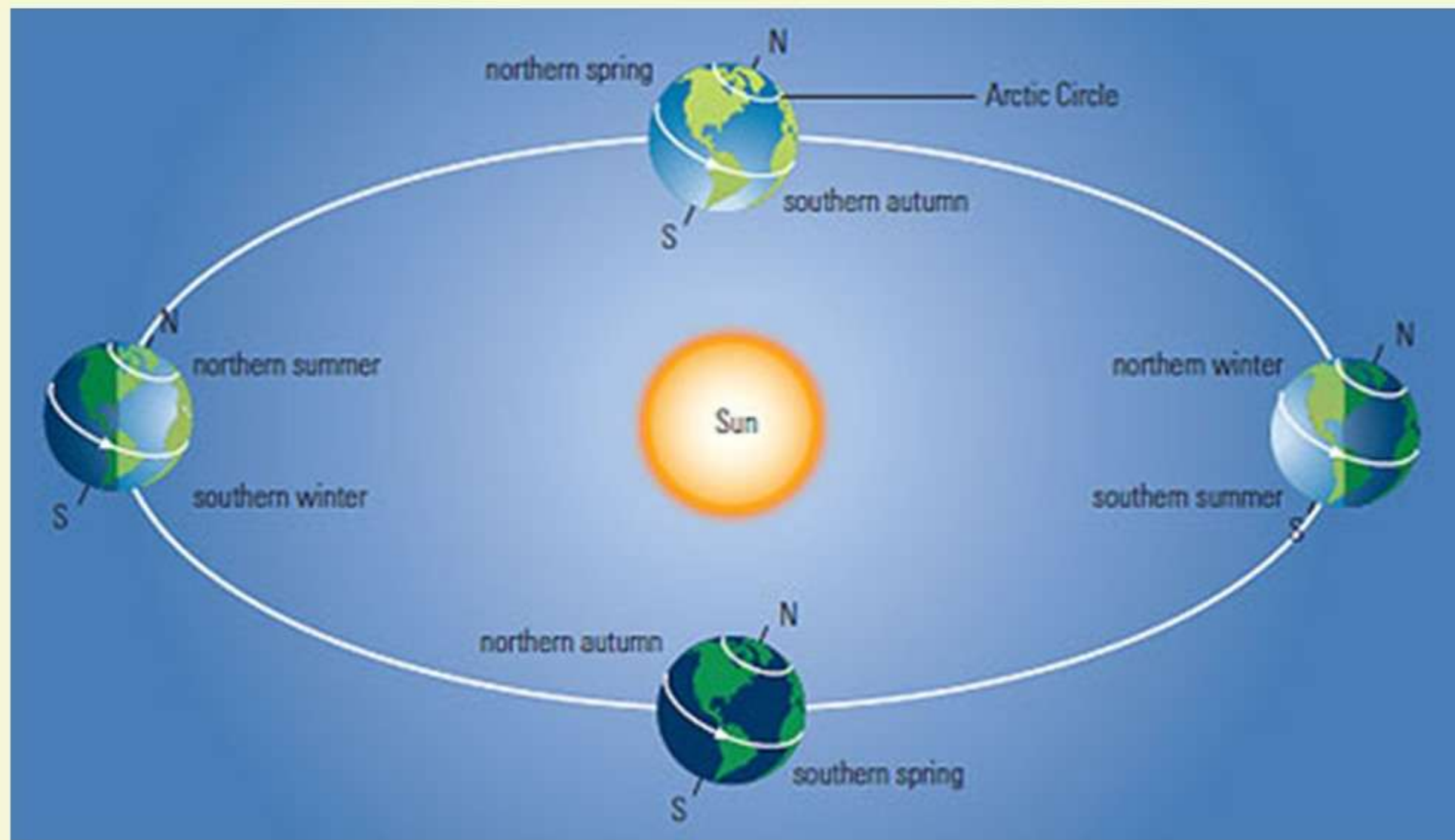
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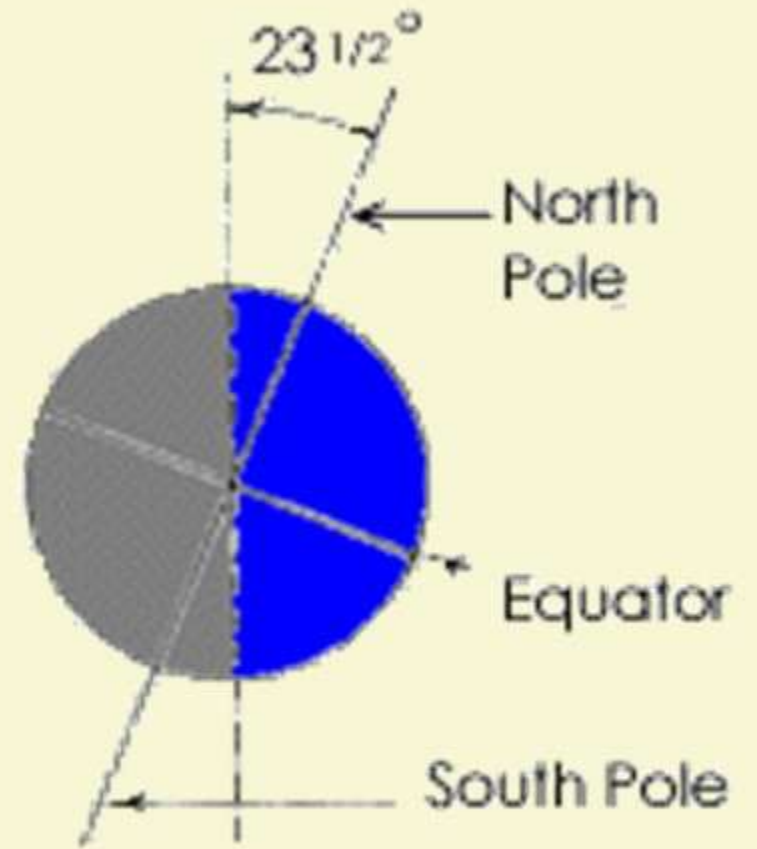
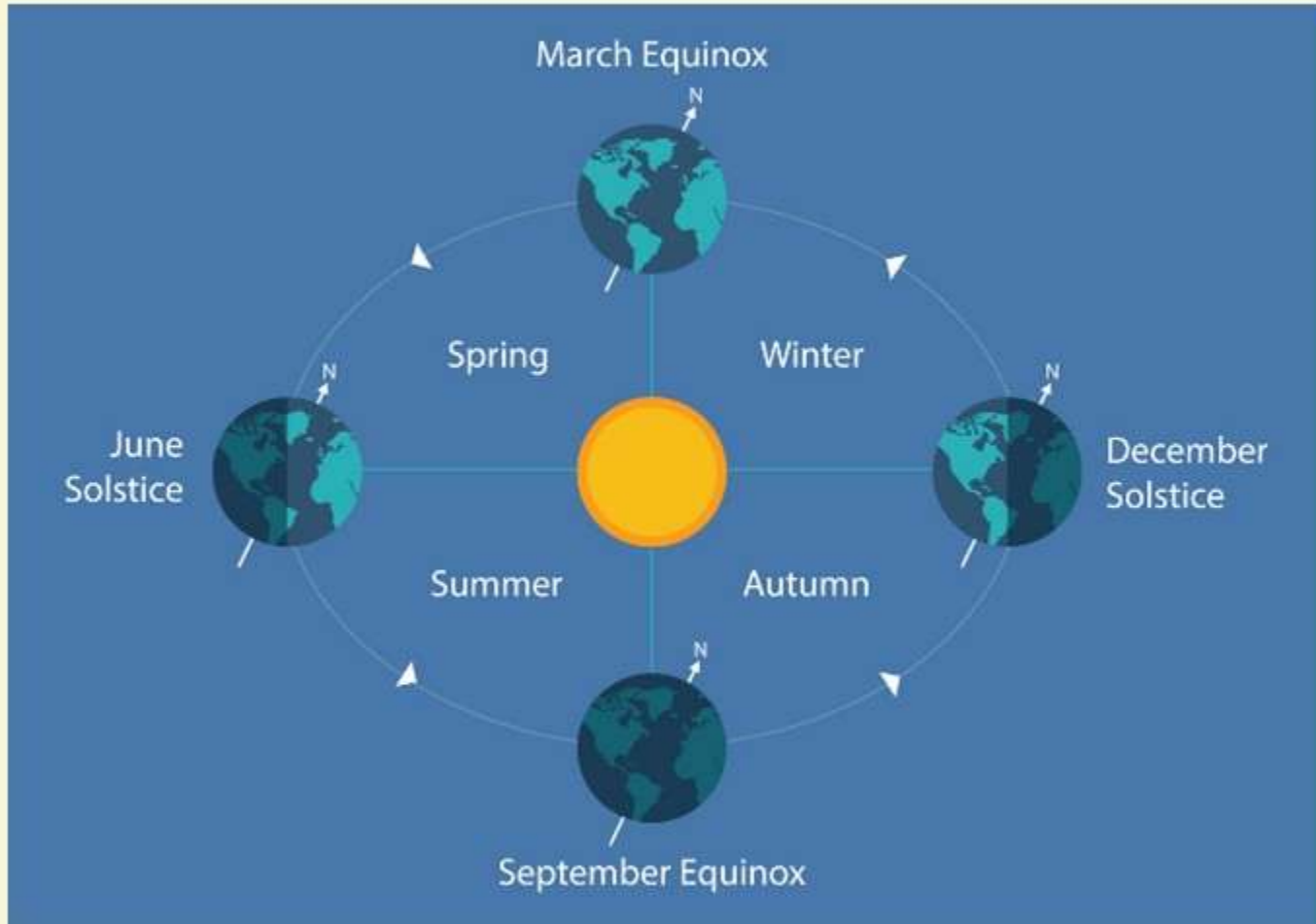


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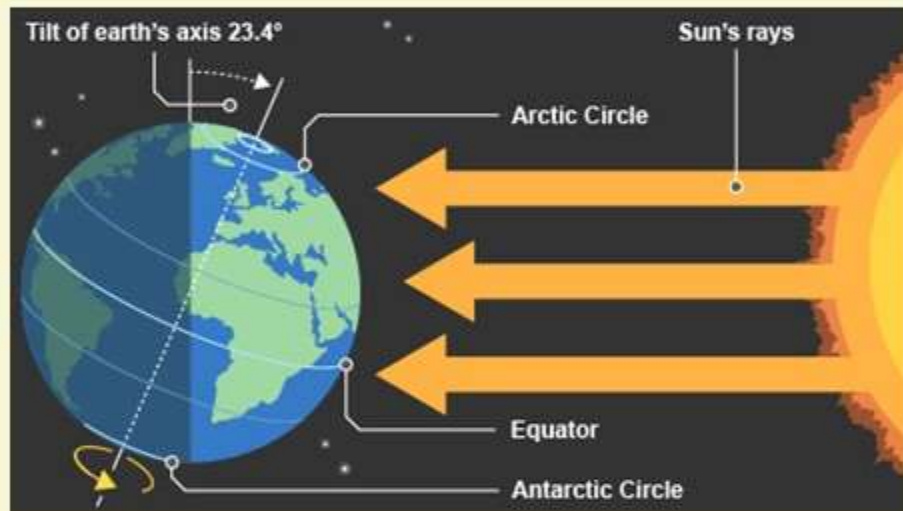
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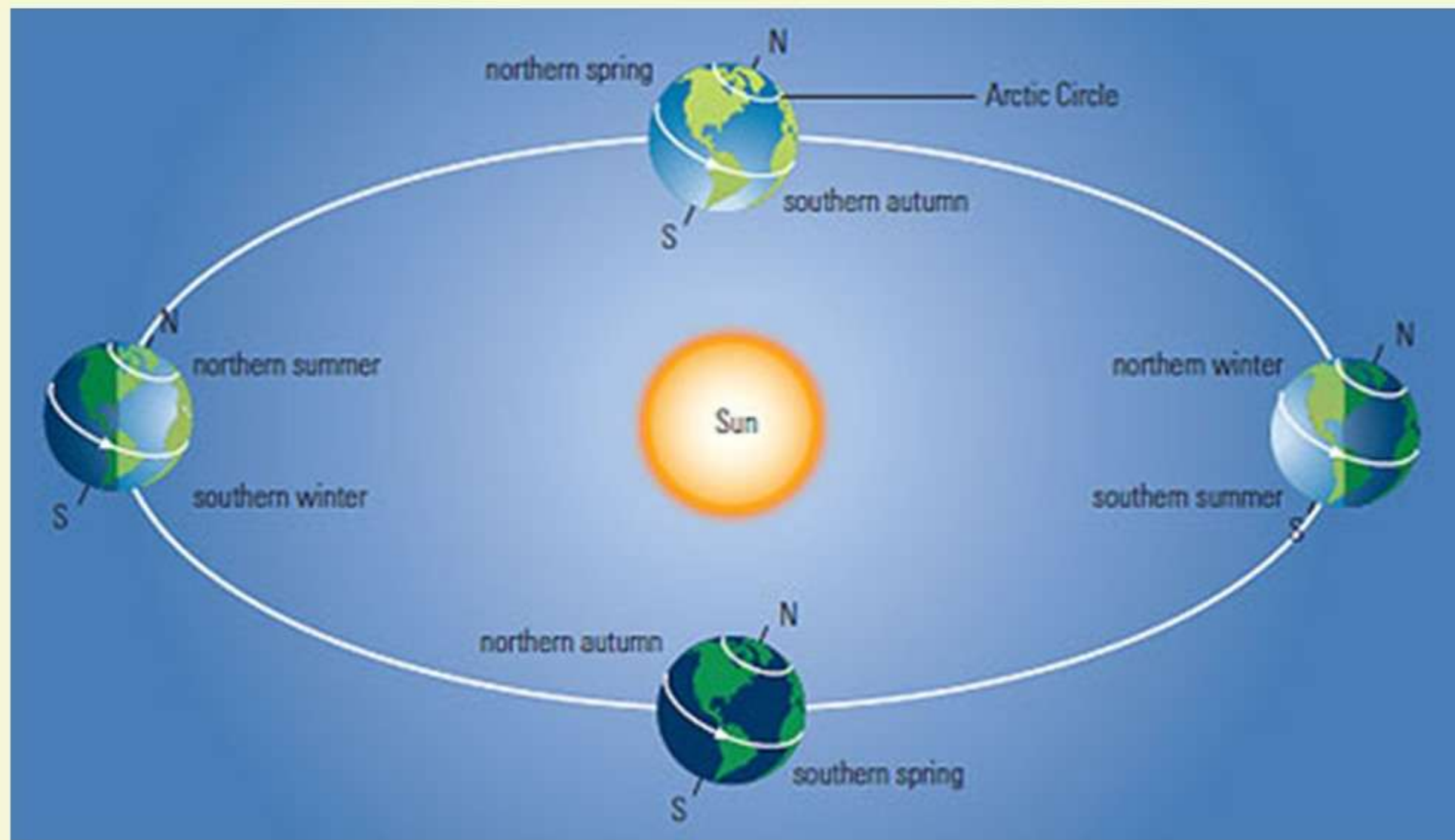


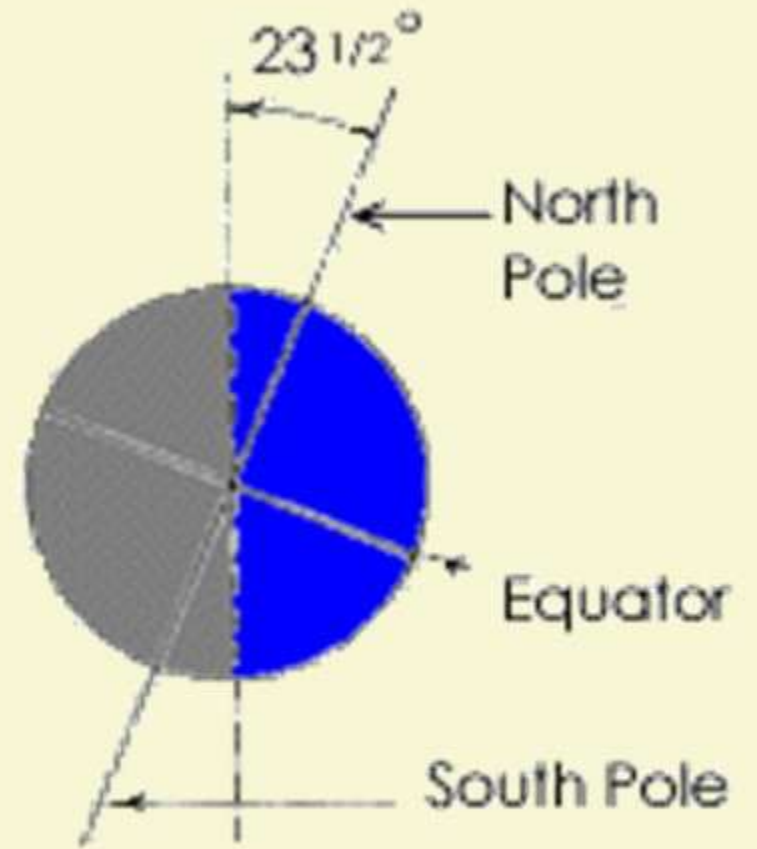
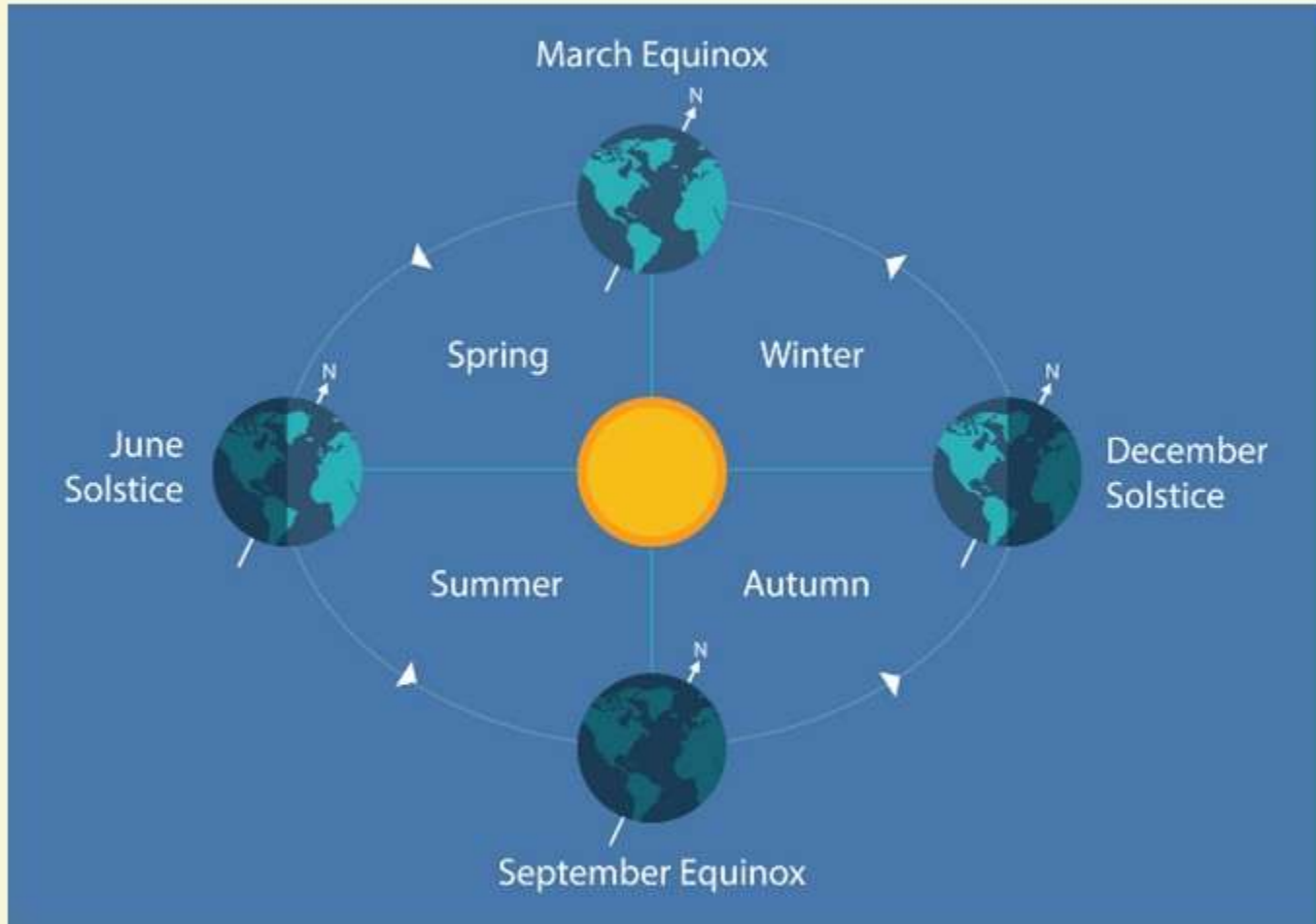


EARTH'S AXIAL TILT



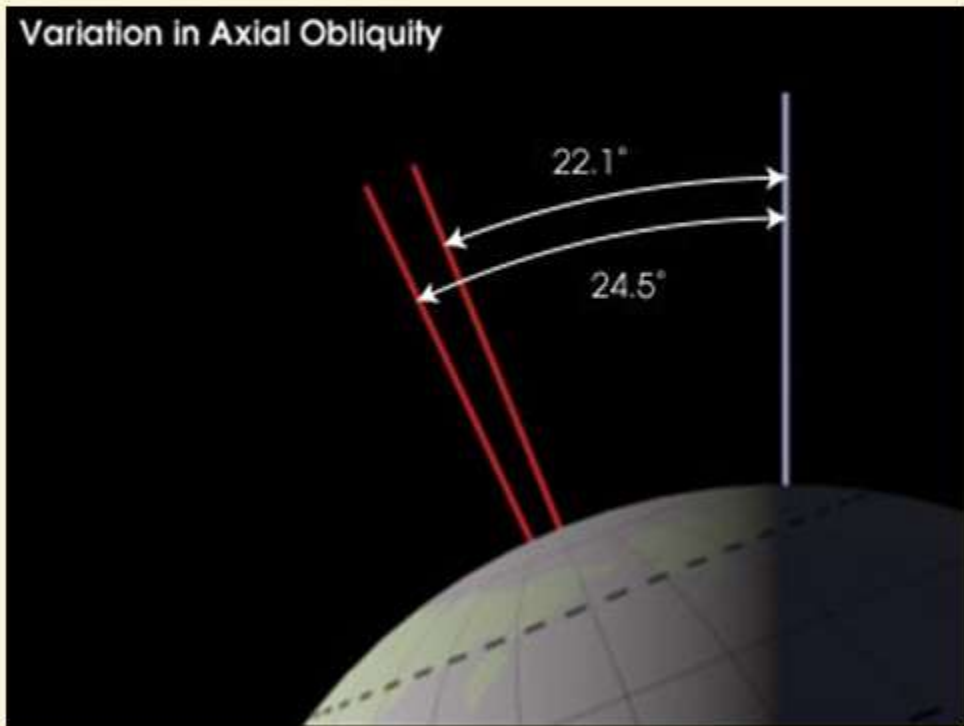
- 1 The ecliptic plane is tilted 23.4° with respect to the plane of the celestial equator.
- 2 Ecliptic is the imaginary plane created by Earth's path around the Sun.
- 3 The planetary bodies of our solar system all tend to lie near this plane.





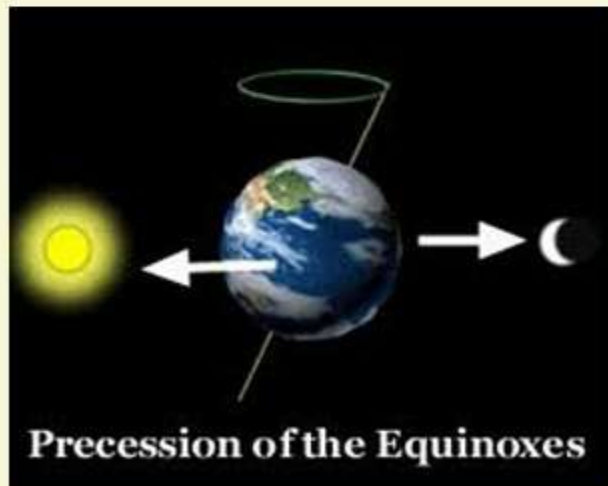
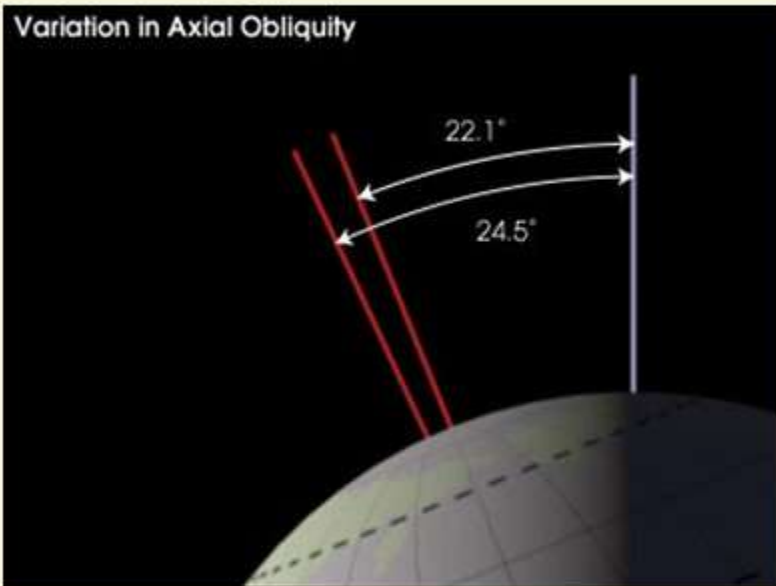
MORE ON EARTH'S AXIAL TILT OR OBLIQUITY

Variation in Axial Obliquity



- 1** Axis is imaginary line that an object rotates around.
- 2** Earth's rotational axis is an imaginary straight line that runs through the North and South poles.
- 3** When an object, the size of Mars crashed into newly formed planet Earth around 4.5 billion years ago, it knocked our planet and left it tilted to an angle.
- 4** For the past 4.5 billion years, Earth is orbiting the Sun at a slant. This is the axial tilt and is called Obliquity

IS THE EARTH'S AXIAL TILT CONSTANT?



- 1 Earth's axial tilt oscillates between 22.1° and 24.5° . The reason is that the Earth's axis also wobbles around itself.
- 2 The wobble motion is called axial precession. It is caused by the gravitational force from the Sun, the Moon, and other planets.
- 3 Axial precession can be described as a slow gyration of Earth's axis about another line intersecting it.
- 4 A complete wobble of Earth's axis takes around 26,000 years.

WHAT HAPPENS BECAUSE OF THIS AXIAL TILT?

1 As the Earth orbits the Sun at an angle, the solar energy reaching different parts of our planet is not constant. It varies during the course of a year.

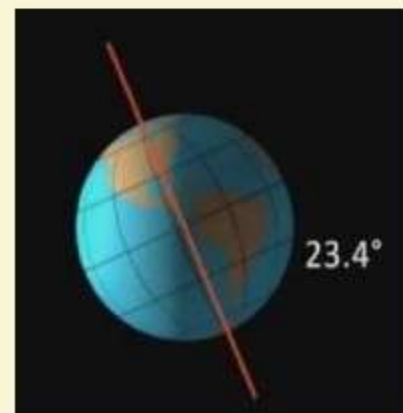
2 This is the reason, why we have different seasons. This is also the reason, why seasons are opposite in the northern and southern hemispheres.

3 The direction of the Earth's tilt nearly does not change.

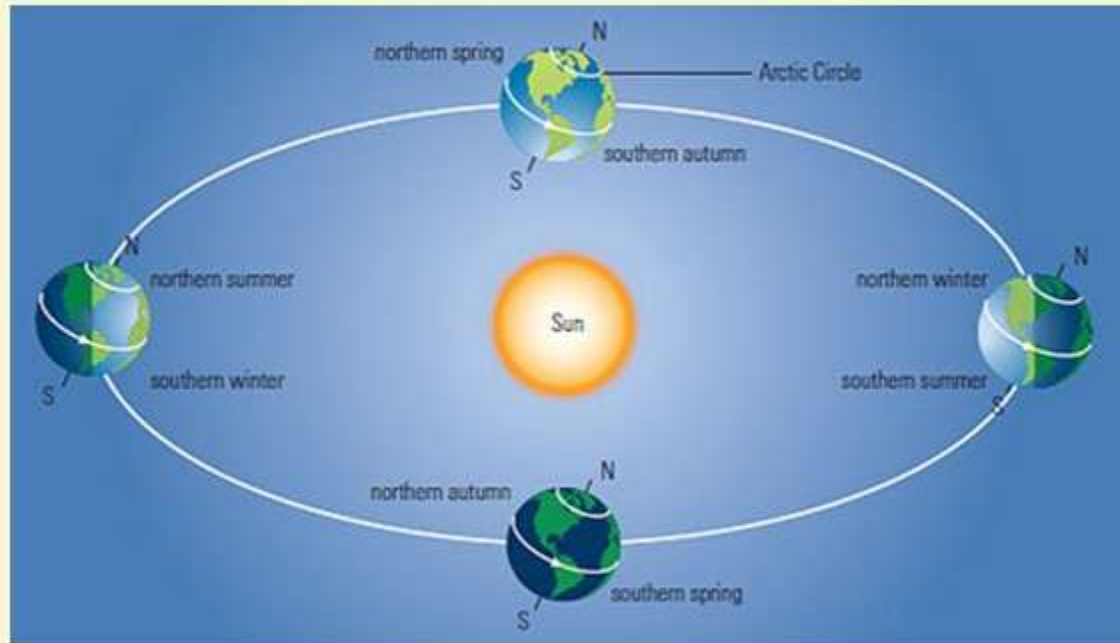
4 The two hemispheres point towards the same position in space through out the entire year.

5 But what does change is, as the Earth revolves around the Sun, the position of the hemisphere in relation to the Sun changes.

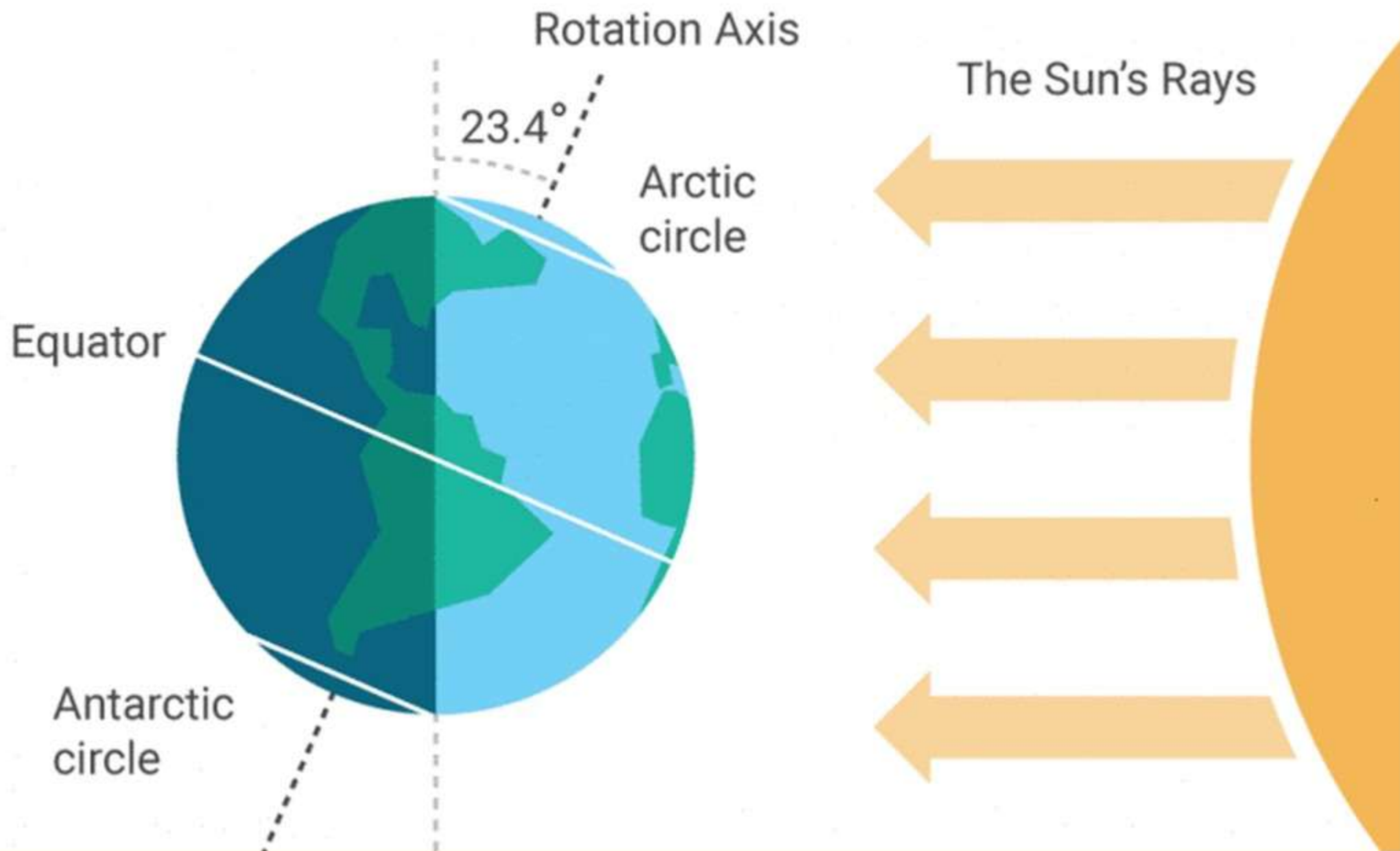
6 Earth's path around the Sun is not circular. Instead, Earth's orbit is elliptical. This means, the Earth's distance from Sun varies through out the year.



SEASONS



- 1 There is common misconception that seasons occur because of the Earth's elliptical orbit around the Sun.
- 2 People feel that winter occurs when Earth is farthest away from the Sun and summer, when it is closest to it.
- 3 However, planet's distance from the Sun has little effect on onset of seasons



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